

# 08 – Numbers & Bases

0 or 1 is not sufficient...

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**DO NOT USE A CALCULATOR!**

0 or 1 is not sufficient...

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# OUTLINE

- 1) Numbers
- 2) Bases
- 3) Binary
- 4) Octal, Hexadecimal and more
- 5) Conversions

# 1) NUMBERS

- In electronic, we use the binary system because it uses two states which are easy to do using electricity
- However, in real life, we use other bases than the base 2 which is not easy for human beings
- Let's have a look at different bases:
  - In base 2 (binary), we have two different symbols to write numbers: 0 and 1
  - In base 10 (decimal), we have . . . different symbols to write numbers: 0 . . .
  - In base 8 (octal ), we have . . . different symbols to write numbers: 0 . . .
  - In base 16 (hexadecimal), we have . . . different symbols to write numbers: 0 . . .

## 2) BASES

- What does it mean to write a number in a certain base?
- What is the number 347 in the decimal base?
- What is the representation of number 347 in base 2 (binary)?
- What is a base?
- Is it possible to use base 3, 4, 5, 6, 7, 9, 11, 12, ...?

## 3) BINARY

- How to find if a number is odd in the binary system?
- How to multiply by two a number in the binary base?
- How to divide by two a number in the binary base?

## 4) OCTAL, HEXADECIMAL AND MORE

- In octal (base 8), there are \_\_\_\_ different symbols:

$$(123)_8 = (\quad)_8$$

$$(456)_8 = (\quad)_8$$

$$(1007)_8 = (\quad)_8$$

$$(104)_{10} = (\quad)_{10}$$

$$(1295)_{10} = (\quad)_{10}$$

$$(4597)_{10} = (\quad)_{10}$$

## 4) OCTAL, HEXADECIMAL AND MORE

- In hexadecimal (base 16), there are \_\_\_\_ different symbols:

$$(12A3)_{16} = ( \quad )_{10}$$

$$(1E0B3)_{16} = ( \quad )_{10}$$

$$(345)_{10} = ( \quad )_{16}$$

$$(67890)_{10} = ( \quad )_{16}$$



## Hexadecimal

## Binary

0

0000

1

0001

2

3

4

5

6

7

8

9

A

B

C

D

E

F

## 4) CONVERSIONS

- Convert:

$$(1\ 0\ 1\ 0\ 1)_2 = (\quad)_{10}$$

$$(1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1)_2 = (\quad)_{10}$$

$$(1\ 0\ 0\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1)_2 = (\quad)_{10}$$

$$(247)_{10} = (\quad)_2$$

$$(1234)_{10} = (\quad)_2$$

$$(10100)_{10} = (\quad)_2$$

## 4) CONVERSIONS

- Convert:
- $(1\ 0\ 1\ 0\ 1)_2 = (\quad)_{16}$
- $(1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1)_2 = (\quad)_{16}$
- $(1\ 0\ 0\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1)_2 = (\quad)_{16}$
- $(2D7)_{16} = (\quad)_2$
- $(12F4)_{16} = (\quad)_2$
- $(10100)_{16} = (\quad)_2$

## 4) CONVERSIONS

- Convert:

$$(1\ 0\ 1\ 0\ 1)_5 = (\quad)_{10}$$

$$(1\ 1\ 0\ 1\ 0\ 1)_7 = (\quad)_{10}$$

$$(1481)_9 = (\quad)_{10}$$

$$(2D7)_{16} = (\quad)_5$$

$$(12F4)_{16} = (\quad)_7$$

$$(10100)_{16} = (\quad)_{11}$$

**ANY QUESTIONS?**